

A Survey of Artificial Intelligence and Metaheuristic Techniques for Enhancing IoT Security

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Abstract

The extensive use of the Internet of Things (IoT) in many different fields has brought about important advancements, but it has also created intricate and ever-changing security issues. This study examines cutting-edge approaches to improving IoT security, with an emphasis on combining metaheuristic algorithms and artificial intelligence (AI) techniques. The goal of the project is to find out how learning-based and optimization-driven methods may identify and lessen security risks including DDoS assaults, data breaches, and encryption flaws. Common AI techniques, such as machine learning and deep learning, as well as metaheuristic algorithms like genetic algorithms, particle swarm optimization, ant colony optimization, and grey wolf optimizer, are reviewed in an organized manner. Performance in intrusion detection, feature selection, cryptographic optimization, and threat prediction is also examined for hybrid models that include AI and metaheuristics. To show the usefulness of these methods in actual IoT environments, a number of successful implementations are highlighted. Problems including scalability problems, lack of interpretability, and computational limitations are also covered. Federated learning, explainable AI, and lightweight architectures are highlighted as viable avenues for developing safe, distributed, and intelligent IoT settings in the paper's final section, which also identifies future research priorities.

Key words: IoT security, artificial intelligence, metaheuristic algorithms, intrusion detection, hybrid models, optimization

1. Introduction

By allowing networked systems that can gather, analyze, and share data in real-time, the Internet of Things (IoT) is revolutionizing a number of industries, including healthcare, smart homes, transportation, and industrial automation[1]. Automation, predictive analytics, and intelligent control are made possible by the widespread integration of digital infrastructure and physical objects[2]. However, because of the devices' heterogeneity, limited processing capabilities, and widespread deployment, the growth of IoT has also brought forth serious security and privacy risks[3]. Cyberthreats, including distributed denial of service (DDoS) attacks, data leaks, malware insertion, and unauthorized access, are becoming more common in IoT systems and have the potential to seriously disrupt services or jeopardize private

information[1]. For identifying and stopping such dynamic and complex attacks, traditional security solutions that rely on manually configured firewalls, static rule sets, or signatures are frequently insufficient[3]. Researchers are increasingly using artificial intelligence (AI) approaches, such as machine learning (ML) and deep learning (DL), to solve these issues. These techniques provide automated and adaptive mechanisms for behavior prediction, threat detection, and anomaly identification[1], [4]. AI-based systems can function with little assistance from humans, learn from past data, and adjust to changing assault patterns[3]. Parallel to this, metaheuristic algorithms like Grey Wolf Optimizer (GWO), Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and Genetic Algorithms (GA) have been extensively employed for security-related optimization tasks like feature selection, parameter tuning, and protocol optimization[4]. Because AI and metaheuristics work well together, hybrid models that combine their advantages have been created[4]. Furthermore, new approaches like federated learning, zero-trust architectures, and generative artificial intelligence (GenAI) are being actively investigated to improve IoT security in dispersed, privacy-sensitive settings[2], [3]. This study offers a thorough analysis of AI and metaheuristic techniques in IoT security, which is important given the growing significance of intelligent and optimal security systems. It discusses important issues, evaluates current developments, offers practical applications, and lists unresolved research issues and potential paths forward [1], [4].

2. IoT Security Challenges and the Role of AI

IoT ecosystems are particularly susceptible because of their size, diversity of devices, limited resources, and frequently decentralized design, all of which increase standard security threats [5]. IoT devices are frequently compromised into botnets that overwhelm network services with malicious traffic, exploiting the devices' weak defenses and limited bandwidth [4]. Sensitive information, such as home sensor readings or health measurements, frequently travels across IoT networks with inadequate encryption or antiquated security measures, raising the danger of disclosure. Simplified or antiquated encryption is used due to resource limitations in IoT devices. Devices become susceptible to brute-force assaults and key compromise as a result [6]. Because of their static nature, sluggish response times, and incapacity to scale over large and dynamic device networks, traditional security solutions such as signature-based firewalls perform poorly in this situation. A powerful remedy for these issues is artificial intelligence (AI), especially machine learning (ML) and deep learning (DL). In IoT contexts, deep learning models incorporating metaheuristic feature optimization, such as the ICSSADL-MHOA framework, have shown DDoS detection accuracy of up to 99.37% [4]. Across all benchmarks, an exceptional 99.99% detection accuracy was attained using a sophisticated hybrid model that used Self-Organizing Maps (SOMs), Deep Belief Networks (DBNs), autoencoders, and Particle Swarm Optimization (PSO) [7]. By effectively identifying intrusions and abnormalities, adjusting to the changing nature of threats, and facilitating real-time reaction, AI-enhanced IoT security solutions demonstrate significant advantages over conventional techniques. Deploying lightweight models appropriate for limited devices, maintaining privacy, balancing false-positive rates, and combining AI with other technologies like edge computing and blockchain still present difficulties.

3. Metaheuristic Algorithms in Security Optimization

Traditional deterministic approaches to security optimization frequently lack the scalability and adaptability needed to handle the growing complexity and dynamic nature of Internet of Things (IoT) settings. Metaheuristic algorithms, which are nature-inspired optimization

approaches that are independent of problems, have become effective tools for improving IoT security systems in response to this [8]. These algorithms are especially useful in tasks such as feature selection, hyperparameter tuning, protocol optimization, and anomaly detection, where exhaustive or brute-force methods are computationally infeasible. Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Grey Wolf Optimizer (GWO) are some of the most popular metaheuristic approaches. GA has been successfully used to improve feature sets used in intrusion detection systems (IDSs), lowering redundancy and improving classification accuracy. It is based on the ideas of natural selection [9]. PSO is commonly used in IoT security frameworks for feature selection and hyperparameter tuning. It is based on the social behavior of bird flocks. On benchmark datasets like NSL-KDD and BoT-IoT, PSO-based hybrid models, notably SOM-DBN-Autoencoder combinations, have demonstrated remarkable performance in identifying sophisticated assaults, with some reaching up to 99.99% accuracy [7]. Additionally, PSO has proven effective in selecting optimal parameters for secure communication protocols in IoT networks, improving both latency and robustness in adversarial conditions [10]. Inspired by ants' foraging habits, ACO has been used to high-dimensional IoT datasets to optimize alert levels and feature selection. It can effectively find optimum or almost optimal solutions in difficult decision spaces thanks to its pheromone-based feedback mechanism [11]. The Grey Wolf Optimizer (GWO), which mimics the hunting habits and leadership structure of grey wolves, is another method that is becoming more and more well-liked. GWO has drawn notice for its capacity to swiftly converge with fewer parameters and for striking a balance between the exploration and exploitation stages. According to recent research, GWO improves detection rates and lowers false-positive ratios in IoT-based intrusion detection systems when it is used to adjust the hyperparameters of machine learning classifiers like Random Forest and SVM [12]. When two or more algorithms are merged to take use of their complementary capabilities, hybridization is when metaheuristics truly shine. For example, as compared to individual algorithms, hybrid GA-PSO and GWO-PSO models have shown better performance in IoT security tasks. These hybrid systems reduce overfitting in classification models, speed up convergence, and increase search space coverage [8]. Three key areas of IoT security optimization have seen the effective use of metaheuristic algorithms. First, feature selection: algorithms such as GA, PSO, and ACO lower computing costs and enhance model generalization by choosing the most pertinent qualities from high-dimensional IoT datasets. Second, hyperparameter tuning: metaheuristics are more effective than grid search or random search techniques in fine-tuning learning rates, layer sizes, and activation functions, which are necessary for models like deep neural networks (DNNs) and ensemble classifiers [12]. Third, protocol configuration and scheduling: For edge-based IoT systems, PSO and GWO have been used to optimize data aggregation schedules, routing pathways, and energy usage, resulting in more robust and secure installations [10].

4. Hybrid AI-Metaheuristic Models: A Modern Approach

Numerous recent studies have suggested hybrid architectures that combine deep learning models with metaheuristic algorithms for improved intrusion detection, lightweight encryption, energy-efficient protocols, and dynamic threat prediction in order to capitalize on the complementary strengths of artificial intelligence (AI) and metaheuristic optimization. Compared to approaches that employ either strategy alone, these hybrid systems have demonstrated significant benefits. One prominent example is the ICSSADL-MHOA model, which combines a BiLSTM network based on self-attention with Hunger Games Search (HGS) for hyperparameter tuning and Improved Tuna Swarm Optimization (ITSO) for feature

selection [4]. This hybrid achieved 99.37% detection accuracy on ToN-IoT and Edge-IIoT datasets, demonstrating the effectiveness of jointly optimizing model architecture and parameters through metaheuristics [4], [13]. Similarly, to enhance anomaly detection in IoT networks, Bensaoud and Kalita (2025) suggested a hybrid architecture that combines Particle Swarm Optimization (PSO) with Autoencoders, Deep Belief Networks (DBN), and Self-Organizing Maps (SOM). By effectively choosing features and adjusting network weights, their SOM–DBN–Autoencoder + PSO system produced an exceptional 99.99% accuracy across the NSL-KDD, UNSW-NB15, and CICIOT2023 datasets [7]. HyMD2I is another noteworthy design that combines deep learning for IoT intrusion detection with an undisclosed metaheuristic optimizer. Initial findings show encouraging performance increases in botnet and anomaly detection, despite the lack of implementation specifics [14]. To improve latency, cost, and load balancing in edge-cloud IoT systems, a hybrid Flower Pollination Algorithm–Tabu Search (FPA–TS) was used for energy and protocol optimization [7]. Their architecture demonstrates how metaheuristic models effectively support hybrid systems beyond detection, extending into operational efficiency for distributed IoT systems. Using Zero-Trust Foundation Models (ZTFMs) in conjunction with blockchain connections and federated learning leads to even more innovation. In order to optimize feature selection and model design in safe AI pipelines spread over dispersed IoT devices, this hybrid paradigm blends metaheuristics [2]. Finally, in Software-Defined Networking (SDN)-based IoT, a hybrid CNN-BiGRU model for DDoS detection was introduced that uses Binary Narwhal Optimizer (BNO) for feature selection and Seagull Optimization Algorithm (SOA) for hyperparameter tuning. This approach achieved 99.40% accuracy on SDN datasets [15]. Hybrid AI–metaheuristic approaches constitute a compelling direction for next-generation IoT security frameworks.

5. Implementation and Case Studies

Numerous real-world applications demonstrate how well AI and metaheuristic combinations work to secure IoT systems. For example, they created a CNN-based intrusion detection system (IDS) that greatly reduced false positives in real-world IoT deployments by utilizing data augmentation and regularization techniques. This system achieved a perfect precision score of 1.0 and an average multi-class precision of 82% across more than ten attack types [16]. A well-known hybrid model, ICSSADL-MHOA, combines metaheuristic optimizers—Hunger Games Search (HGS) for hyperparameter tuning and Improved Tuna Swarm Optimization (ITSO) for feature selection with self-attention BiLSTM detection. This system's 99.37% detection accuracy on the ToN-IoT and Edge-IIoT datasets confirmed its suitability for use in extensive IoT networks [4]. presented a SOM–DBN–Autoencoder IDS model that was improved by Particle Swarm Optimization (PSO) in a different instance. The model demonstrated strong identification across several IoT threat landscapes with a high Matthews Correlation Coefficient (MCC) and 99.99% accuracy when tested on the NSL-KDD, UNSW-NB15, and CICIOT2023 datasets [7]. The optimization of edge-cloud resources is also covered in case studies. Optimized resource allocation across edge and cloud infrastructures by combining the Flower Pollination Algorithm with Tabu Search (FPA–TS). The technique decreased job duration and cost in simulations while preserving QoS, which is a useful strategy that indirectly improves security by boosting system resilience [17]. Additionally, real-world models that combine hybrid metaheuristics and deep learning have demonstrated success in SDN-enabled IoT environments. A recent study detected DDoS attacks in software-defined networks with 99.40% accuracy by fusing CNN and BiGRU

architectures with Binary Narwhal Optimization (BNO) and Seagull Optimization Algorithm (SOA) for feature and parameter selection [4].

6. Challenges, Limitations, and Future Trends

There are still a number of technological obstacles in protecting IoT settings, even with significant progress in AI-metaheuristic hybrid models. First and foremost, the deployment of complicated detection models on IoT devices is still hampered by resource limitations, such as restricted memory, processing power, and energy, which also limit real-time performance and power efficiency [8]. Second, model learning is greatly impacted by the imbalance and variety of IoT datasets: unusual attack classes are frequently underrepresented, which results in poor generalization and high false-negative rates [18]. Explainability and trustworthiness pose further obstacles: complex deep learning and metaheuristic combinations often behave as “black-boxes,” making it difficult for stakeholders to understand decision logic, especially in regulated domains like healthcare or industrial automation [19]. Furthermore, when metaheuristic parameters are poorly described or adjusted on small datasets, repeatability and overfitting continue to be issues, restricting application across different IoT platforms and devices [20]. On-device learning is made possible via Federated Learning (FL) and Decentralized Intelligence frameworks, which also reduce bandwidth and privacy issues. By facilitating fidelity-calibrated IDS through secure FL, methods such as Choir-IDS lessen the need for central data aggregation while maintaining model performance [19], [21]. By combining curriculum learning with explainable AI (XAI), model robustness and transparency may be increased. A lightweight curriculum-learning + XAI system, for instance, enabled feature-level explanations via LIME and obtained around 98% accuracy on CIC-IoV-2024 and EDGE-IIoT [22]. CNN/RNN models and knowledge graphs are used in semantic reasoning-enhanced deep learning systems to provide interpretability and context-aware threat identification. Scalability, GDPR/CCPA compliance, and great flexibility were proven by a 2025 system on CICIoT2023 datasets [23]. Class-balanced optimization and adaptive energy usage are key components of next-generation metaheuristics designed for IoT efficiency and fairness. Better convergence with fewer nodes and energy burden is promised by novel algorithms that unite WOA–GWO hybrids [8], [24]. Integration of generative AI (GenAI) into IDS pipelines is becoming a new frontier. Although there are still ethical and antagonistic hazards, recent surveys indicate that GenAI can improve anomaly detection training, mimic attack situations, and assist in predicting new threat patterns [3], [25]. To allow repeatable evaluation of AI and metaheuristic hybrids, standardization and a variety of benchmark datasets are desperately needed, especially ones that include zero-day assaults and realistic IoT deployment situations [26]. When implementing AI on limited IoT devices, edge-aware energy-efficient learning techniques like quantization, pruning, and lightweight model design assist lower runtime and power costs [22], [23].

7. Conclusion and Research Directions

We have examined in this survey how metaheuristic methods and artificial intelligence (AI) are changing the IoT security environment. We started by recognizing the increasing problems that IoT systems face, which are made worse by the size and diversity of contemporary IoT installations. These problems include DDoS assaults, data leaks, and inadequate encryption techniques. These problems necessitate resource-efficient, flexible, and intelligent security solutions. Deep learning, reinforcement learning, and behavior-based analysis are a few examples of AI approaches that have shown remarkable efficacy in tasks including threat prediction, anomaly detection, and intrusion detection. When combined with

real-time data from various IoT nodes, these techniques allow for dynamic responses to changing threats. In the meantime, metaheuristic algorithms like Grey Wolf Optimizer (GWO), Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and Genetic Algorithms (GA) have proven to be effective in feature selection, security-related parameter optimization, and setting up lightweight cryptographic schemes. Our examination of hybrid AI-metaheuristic models demonstrates the significant benefits that come from combining them: While metaheuristics aid in the exploration and optimization of the solution space, artificial intelligence (AI) offers learning and prediction capabilities. These hybrid systems may be used for threat forecasting, energy-efficient and secure intrusion detection, lightweight encryption techniques, and extensive data processing in Internet of Things settings. These systems are scalable to the limitations of edge and fog computing architectures while continuously achieving high detection accuracy and performance adaptation. Notwithstanding these successes, difficulties still exist. Lack of explainability in black-box models, data quality problems, limited processing capacity on edge devices, and the lack of standardized, heterogeneous IoT datasets frequently impede real-world deployment. Furthermore, static defenses lose their effectiveness as opponents get more complex. A number of study avenues become apparent as we look to the future. Privacy-preserving model training over dispersed nodes can be supported by decentralized techniques like federated learning. Explainable AI will be essential for fostering openness and trust, especially in fields where safety is a top priority. Methods of curriculum-based learning can improve resilience and generalization, particularly in dynamic threat contexts. Furthermore, reducing latency and power consumption in edge-based deployments will need the use of hybrid algorithms and novel energy-aware optimization techniques. Adaptive, interpretable, and efficient intelligent systems built to not only identify and address present risks but also predict and neutralize new ones in real time are essential to the future of IoT security. Continued cooperation across the AI, cybersecurity, and embedded systems groups is necessary to realize this goal, with a focus on reliable, explicable, and scalable solutions that are adapted to the ever-changing IoT ecosystems.

References

- [1] M. S. Ahsan, S. Islam, and S. Shatabda, "A Systematic Review of Metaheuristics-Based and Machine Learning-Driven Intrusion Detection Systems in IoT," *Swarm and Evolutionary Computation*, vol. 96, p. 101984, Jul. 2025, doi: 10.1016/j.swevo.2025.101984.
- [2] K. Li *et al.*, "Zero-Trust Foundation Models: A New Paradigm for Secure and Collaborative Artificial Intelligence for Internet of Things," May 26, 2025, *arXiv*: arXiv:2505.23792. doi: 10.48550/arXiv.2505.23792.
- [3] Y. L. Aung, I. Christian, Y. Dong, X. Ye, S. Chattopadhyay, and J. Zhou, "Generative AI for Internet of Things Security: Challenges and Opportunities," Feb. 13, 2025, *arXiv*: arXiv:2502.08886. doi: 10.48550/arXiv.2502.08886.
- [4] F. Alblehai, "Artificial intelligence-driven cybersecurity system for internet of things using self-attention deep learning and metaheuristic algorithms," *Sci Rep*, vol. 15, no. 1, p. 13215, Apr. 2025, doi: 10.1038/s41598-025-98056-2.
- [5] A. Alfahaid, E. Alalwany, A. M. Almars, F. Alharbi, E. Atlam, and I. Mahgoub, "Machine Learning-Based Security Solutions for IoT Networks: A Comprehensive Survey," *Sensors*, vol. 25, no. 11, Art. no. 11, Jan. 2025, doi: 10.3390/s25113341.
- [6] A. Pakmehr, A. Aßmuth, N. Taheri, and A. Ghaffari, "DDoS attack detection techniques in IoT networks: a survey," *Cluster Comput*, vol. 27, no. 10, pp. 14637–14668, Dec. 2024, doi: 10.1007/s10586-024-04662-6.

- [7] A. Bensaoud and J. Kalita, "Optimized detection of cyber-attacks on IoT networks via hybrid deep learning models," *Ad Hoc Networks*, vol. 170, p. 103770, Apr. 2025, doi: 10.1016/j.adhoc.2025.103770.
- [8] M. S. Ahsan, S. Islam, and S. Shatabda, "A Systematic Review of Metaheuristics-Based and Machine Learning-Driven Intrusion Detection Systems in IoT," *Swarm and Evolutionary Computation*, vol. 96, p. 101984, Jul. 2025, doi: 10.1016/j.swevo.2025.101984.
- [9] H. Ma, W. Zhang, D. Zhang, and B. Chen, "An IoT intrusion detection framework based on feature selection and large language models fine-tuning," *Sci Rep*, vol. 15, no. 1, p. 21158, Jul. 2025, doi: 10.1038/s41598-025-08905-3.
- [10] J. García-Nieto, J. Toutouh, and E. Alba, "Automatic tuning of communication protocols for vehicular ad hoc networks using metaheuristics," *Engineering Applications of Artificial Intelligence*, vol. 23, no. 5, pp. 795–805, Aug. 2010, doi: 10.1016/j.engappai.2010.01.012.
- [11] K. Seo, H. H. Refai, and E. S. Hile, "Application of Dynamic Mode Decomposition to Characterize Temporal Evolution of Plantar Pressures from Walkway Sensor Data in Women with Cancer," *Sensors*, vol. 24, no. 2, Art. no. 2, Jan. 2024, doi: 10.3390/s24020486.
- [12] "A metaheuristic-based ensemble feature selection framework for cyber threat detection in IoT-enabled networks," *Decision Analytics Journal*, vol. 7, p. 100206, Jun. 2023, doi: 10.1016/j.dajour.2023.100206.
- [13] "(PDF) Hybrid Metaheuristic and Deep Learning Models for Intrusion Detection in Cybersecurity Networks," ResearchGate. Accessed: Jul. 10, 2025. [Online]. Available: https://www.researchgate.net/publication/393446955_Hybrid_Metaheuristic_and_Deep_Learning_Models_for_Intrusion_Detection_in_Cybersecurity_Networks
- [14] "HyMD2I: Hybrid Metaheuristic-Deep Learning Approach for Intrusion Detection in IoT | Request PDF," *ResearchGate*, Jul. 2025, doi: 10.1007/s10922-025-09946-9.
- [15] C. L. Kumar *et al.*, "Metaparameter optimized hybrid deep learning model for next generation cybersecurity in software defined networking environment," *Sci Rep*, vol. 15, no. 1, p. 14166, Apr. 2025, doi: 10.1038/s41598-025-96153-w.
- [16] M. S. Amine, F. A. Nada, and K. M. Hosny, "Improved model for intrusion detection in the Internet of Things," *Sci Rep*, vol. 15, no. 1, p. 21547, Jul. 2025, doi: 10.1038/s41598-025-92852-6.
- [17] N. M. Dankolo *et al.*, "Optimizing resource allocation for IoT applications in the edge cloud continuum using hybrid metaheuristic algorithms," *Sci Rep*, vol. 15, no. 1, p. 14409, Apr. 2025, doi: 10.1038/s41598-025-97648-2.
- [18] S. Jamshidi, A. Nikanjam, N. K. Wazed, and F. Khomh, "Leveraging Machine Learning Techniques in Intrusion Detection Systems for Internet of Things," Apr. 09, 2025, *arXiv*: arXiv:2504.07220. doi: 10.48550/arXiv.2504.07220.
- [19] E. Dritsas and M. Trigka, "Federated Learning for IoT: A Survey of Techniques, Challenges, and Applications," *Journal of Sensor and Actuator Networks*, vol. 14, no. 1, Art. no. 1, Feb. 2025, doi: 10.3390/jsan14010009.
- [20] M. Rahmati, "Federated Learning-Driven Cybersecurity Framework for IoT Networks with Privacy-Preserving and Real-Time Threat Detection Capabilities," Feb. 14, 2025, *arXiv*: arXiv:2502.10599. doi: 10.48550/arXiv.2502.10599.
- [21] "Choir-IDS: A federated learning framework for fidelity-calibrated explainable intrusion detection system for edge-IoT networks," *Information Fusion*, vol. 125, p. 103473, Jan. 2026, doi: 10.1016/j.inffus.2025.103473.
- [22] S. Narkedimilli, S. Makam, A. V. Sriram, S. P. Mallellu, M. Sathvik, and R. R. V. Prasad, "Enhancing IoT Network Security through Adaptive Curriculum Learning and XAI," Jan. 20, 2025, *arXiv*: arXiv:2501.11618. doi: 10.48550/arXiv.2501.11618.
- [23] R. Ranpara, S. K. Patel, O. P. Kumar, and F. A. Al-Zahrani, "A computational framework for IoT security integrating deep learning-based semantic algorithms for real-time threat response," *Sci Rep*, vol. 15, no. 1, p. 16794, May 2025, doi: 10.1038/s41598-025-93898-2.

- [24]“A systematic review of metaheuristics-based and machine learning-driven intrusion detection systems in IoT,” *Swarm and Evolutionary Computation*, vol. 96, p. 101984, Jul. 2025, doi: 10.1016/j.swevo.2025.101984.
- [25]J. L. López Delgado and J. A. López Ramos, “A Comprehensive Survey on Generative AI Solutions in IoT Security,” *Electronics*, vol. 13, no. 24, Art. no. 24, Jan. 2024, doi: 10.3390/electronics13244965.
- [26]W. Fei, H. Ohno, and S. Sampalli, “A Systematic Review of IoT Security: Research Potential, Challenges, and Future Directions,” *ACM Comput. Surv.*, vol. 56, no. 5, p. 111:1-111:40, Nov. 2023, doi: 10.1145/3625094.